HAMDA HMIDA

AI Research Engineer

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in hamda-hmida HamdaHmida



PROFESSIONAL EXPERIENCES

AI Researcher

CEMEF, Mines Paris - PSL

Dec 2024 – Nov 2027 Sophia Antipolis

- Advancing Foundation Models for fluid flow simulation and prediction.

Graduation Internship

AIRBUS

Mar 2024 – Sep 2024 Toulouse

- Design a Reinforcement Learning algorithm to assist pilots in emergency diversions by optimizing for minimal time, fuel consumption, and diversion cost, with the goal of reaching the nearest airport with explanations.

Software Engineer

ReDX Technologies

Jun 2023 – Jan 2024 Sfax

- Conduct in-depth research to optimize the Digital Rock Physics Workflow using super resolution, diffusion models, and segmentation techniques.
- Investigate the application of MONAI, a medical imaging deep learning framework, to analyze digital rock images for oil exploration.

ACADEMIC BACKGROUND

DOCTOR OF PHILOSOPHY

École des Mines de Paris , PSL University

Dec 2024 – Nov 2027

Doctoral studies in Numerical Mathematics, High-Performance Computing, and Data. PhD Defense will take place in November 2027

NATIONAL ENGINEERING DEGREE

Ecole Polytechnique de Tunisie , University of Carthage

Sep 2021 – Sep 2024

Multidisciplinary Engineering Studies - Master of Engineering Majoring in Signals and Systems

PRE-ENGINEERING CYCLE

The Preparatory Institute for Engineering Studies of Monastir

Sep 2019 – Jun 2021

Finished by succeeding in the national pre-engineering exam (Physics - Chemistry) with a national rank : 8 / 1000

RESEARCH PAPERS

- Hamda Hmida, Hsiu-Wen Chang Joly, and Youssef Mesri. "Compositional Neural Operators for Multi-Dimensional Fluid Dynamics", AI&PDE: ICLR 2026 Workshop on AI and Partial Differential Equations.
- Hamda Hmida, Hsiu-Wen Chang Joly, and Youssef Mesri. "CompNO: A novel foundation model approach for solving partial differential equations". Applied Sciences, 16(2), 2026. ISSN 2076-3417.

SUMMARY

Highly motivated and results-oriented, passionate about continuous learning and development and actively seeking to contribute significantly to the team's success

CERTIFICATES

Neural Networks and Deep Learning
Machine Learning with Python

HONORS & AWARDS

Computer Vision for License Plate Recognition, **Ranked 1/88** (Zindi)
UmojaHack2023 Cryptojacking Challenge, **Ranked 3/247** (Zindi)
Tunisian Collegiate Programming Contest, **Ranked 16/120**

PROGRAMMING AND IT TOOLS

Python | PyTorch | C/C++ | Java | CUDA | Bash Scripting | Git

TECH KNOWLEDGE

Data analytics | Machine Learning | Reinforcement learning | Planning | Programming | Problem Solving

SOFT SKILLS

Teamwork | Communication | Leadership | Time management

LANGUAGES

English (TOEIC B2)
French (DELF B2)
Arabic : Native

PASSION

Traveling | Video Games | Ping-Pong